```python
import tkinter as tk
from firebase_admin import db

from learning_module import LearningModule, Lesson, Quiz
from users.user import User


class Learner(User):
    """
    This is the class definition for the Learner class.
    """

    def __init__(self, email, first_name, last_name, username, password,
    date_of_birth):
        """
        Constructor for the Learner class.
        """
        super().__init__(
            email, first_name, last_name, username, password, role="Learner"
        )
        self.date_of_birth = date_of_birth
        self.achievements = []
        self.modules = []

    def populate_modules(self) -> list[LearningModule]:
        """
        This method is used to populate the modules from the database.

        :return: A list of LearningModule objects.
        """
        ref = db.reference("/modules")
        modules = []
        modules_data = ref.get()
        for module in modules_data:
            # Convert lessons and quizzes to Lesson and Quiz objects
            lessons = []
            for lesson_data in module["lessons"]:
                lessons.append(Lesson(lesson_data["title"], lesson_data["content"]))

            quizzes = []
            for quiz_data in module["quizzes"]:
                quizzes.append(
                    Quiz(
                        quiz_data["question"],
                        quiz_data["options"],
                        quiz_data["answer"],
                        quiz_data["hint"],
                        quiz_data["solution"],
                    )
                )

            # Create LearningModule object
            modules.append(
                LearningModule(
                    module["title"],
                    module["description"],
                    lessons,
                    quizzes,
                    module["difficulty"],
                    module["achievement"],
                )
            )

        return modules

    def dashboard(self, login_window=None):
        """
        This method is used to display the dashboard for the learner.

        :param login_window: The login window to be destroyed after successful login.
        """
        if login_window:
            login_window.destroy()
```

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73
74          self.modules = self.populate_modules()
75
76          self.dashboard_window = tk.Tk()
77          self.dashboard_window.title("Learner Dashboard")
78
79          welcome_label = tk.Label(
80              self.dashboard_window, text=f"Welcome, {self.first_name}!"
81          )
82          welcome_label.pack(pady=10)
83
84          self.init_buttons()
85
86          # while True:
87          #     print(f"\nWelcome {self.first_name}\n")
88          #     print("1. Learning Modules")
89          #     print("2. Achievements")
90          #     print("3. Edit Profile")
91          #     print("4. FAQ")
92          #     print("5. Logout")
93          #     choice = input("Enter your choice: ")
94
95          #     if choice == "1":
96          #         self.view_learning_modules()
97          #     elif choice == "2":
98          #         self.view_achievements()
99          #     elif choice == "3":
100         #         self.edit_profile()
101         #     elif choice == "4":
102         #         super().view_faq()
103         #     elif choice == "5":
104         #         break
105         #     else:
106         #         print("Please choose a valid option.")
107
108     def init_buttons(self):
109         """
110         This method is used to initialize the buttons for the learner dashboard in
            tkinter.
111         """
112         modules_button = tk.Button(
113             self.dashboard_window,
114             text="View Learning Modules",
115             command=self.view_learning_modules,
116         )
117         modules_button.pack(pady=10)
118         modules_button = tk.Button(
119             self.dashboard_window,
120             text="View Achievements",
121             command=self.view_achievements,
122         )
123         modules_button.pack(pady=10)
124
125         # TODO: theres a runtime error for some reason here after pressing the button
126         modules_button = tk.Button(
127             self.dashboard_window, text="Edit Profile", command=self.edit_profile
128         )
129         modules_button.pack(pady=10)
130
131     def view_learning_modules(self):
132         """
133         This method is used to display the learning modules for the learner.
134         """
135         print("\n====== Modules ======\n")
136         for i in range(len(self.modules)):
137             print(
138                 f"{i+1}. {self.modules[i].get_title()} -
                    {self.modules[i].get_description()}\n"
139             )
140         print("====================\n")
141         print("1. View a module")
142         print("2. Go back")
143
```

```
144             choice = input("Enter your choice: ")
145             if choice == "1":
146                 module_choice = int(input("Enter module number: "))
147                 self.modules[module_choice - 1].view_module()
148             elif choice == "2":
149                 return
150             else:
151                 print("Please choose a valid option.")
152 
153         def view_module(self, module: LearningModule):
154             """
155             This method is used to display a learning module for the learner.
156 
157             :param module: The learning module to be displayed.
158             """
159             print("\n====== Module ======\n")
160             print(f"{module['title']}\n")
161             print(f"{module['description']}\n")
162             print("Lessons:")
163             for lesson_data in module["lessons"]:
164                 lesson = Lesson(lesson_data["title"], lesson_data["content"])
165                 lesson.view_lesson()
166             print("\nQuizzes:")
167             for quiz_data in module["quizzes"]:
168                 quiz = Quiz(
169                     quiz_data["question"],
170                     quiz_data["options"],
171                     quiz_data["answer"],
172                     quiz_data["hint"],
173                     quiz_data["solution"],
174                 )
175                 # quiz.attempt_quiz()
176             print("\n====================\n")
177             print("1. View a lesson")
178             print("2. View a quiz")
179             print("3. Go back")
180 
181             choice = input("Enter your choice: ")
182             if choice == "1":
183                 lesson_choice = int(input("Enter lesson number: "))
184                 self.view_lesson(module, lesson_choice)
185             elif choice == "2":
186                 quiz_choice = int(input("Enter quiz number: "))
187                 self.view_quiz(module, quiz_choice)
188             elif choice == "3":
189                 return
190             else:
191                 print("Please choose a valid option.")
192 
193         def view_achievements(self):
194             """
195             This method is used to display the achievements for the learner.
196             """
197             print("\n🏆 Achievements 🏆\n")
198             for achievement in self.achievements:
199                 print(achievement)
200 
```